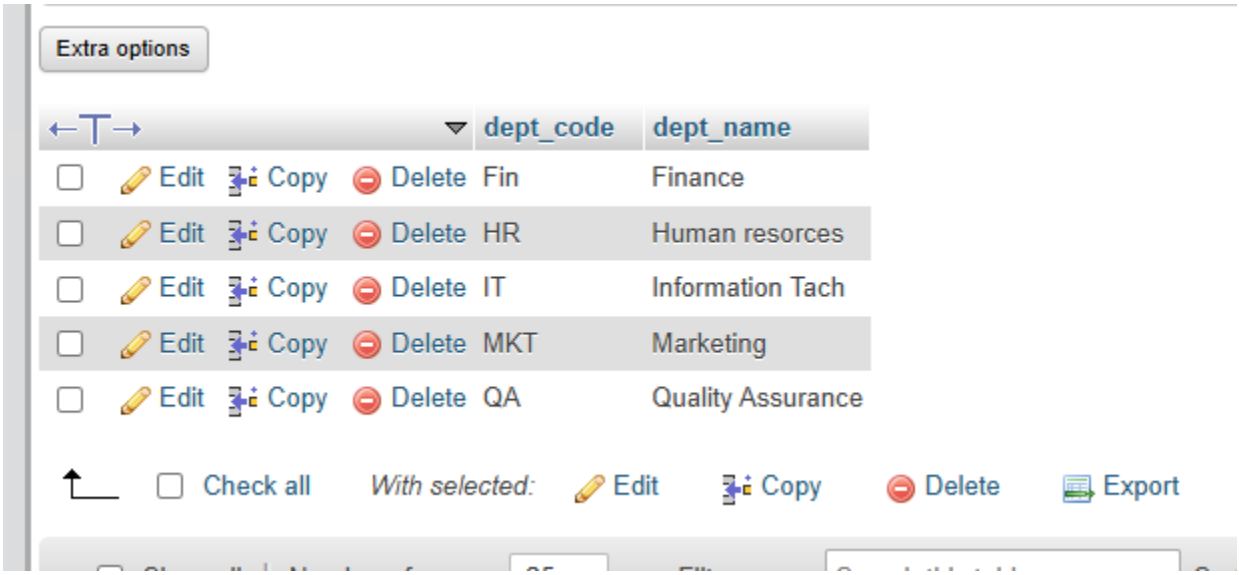
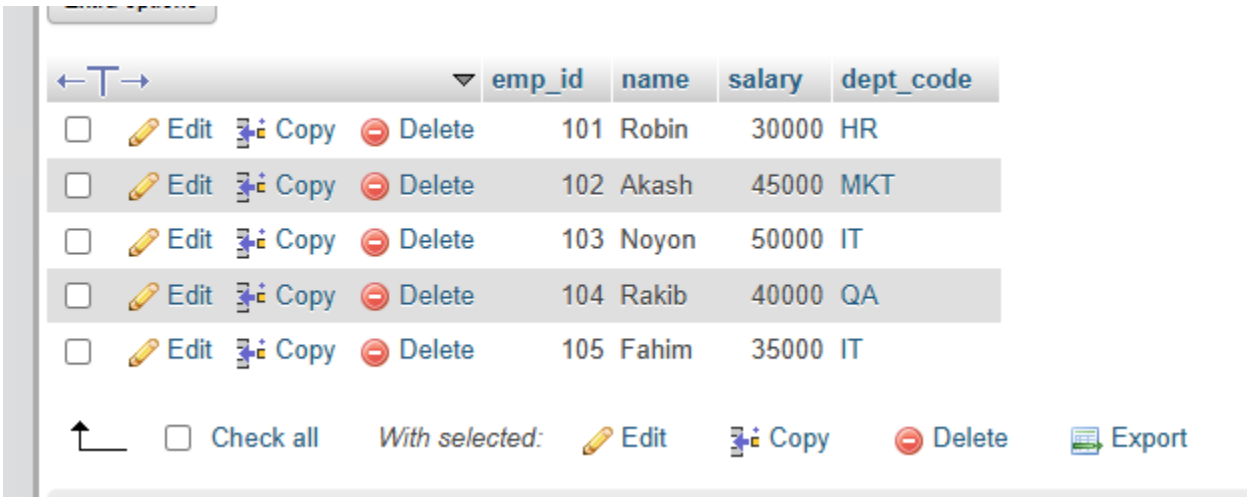
VIEW

_____





CREATE VIEW emp_view as

select name, salary from employee

select name from employee

theory:

NO Memory needs for virtual table. It finds by execution

view with another view

CREATE view emp_view_name as

SELECT name from emp_view

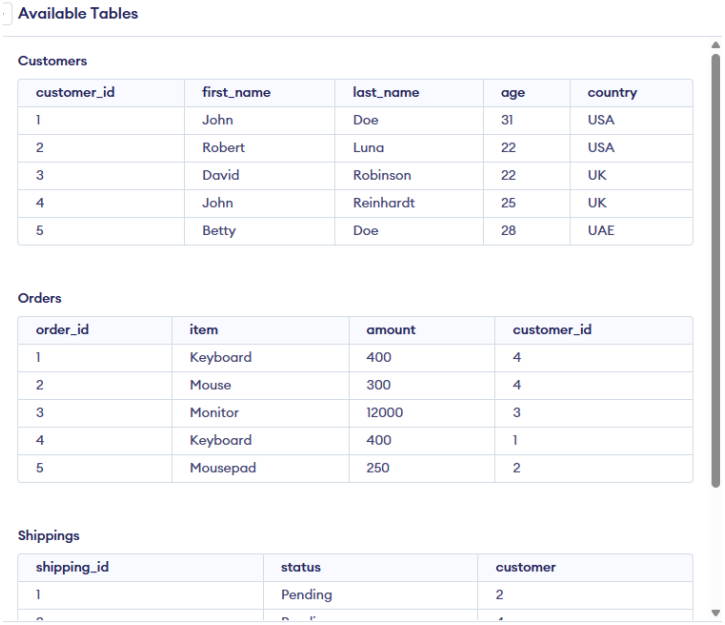
Create view with join two table for show datas from different table

---

CREATE VIEW emp_dept as

select e.name, d.dept_code, d.dept_name from employee e INNER JOIN department d on e.dept_code = d.dept_code

Limit



Available Tables

Customers

customer_id	first_name	last_name	age	country
1	John	Doe	31	USA
2	Robert	Luna	22	USA
3	David	Robinson	22	UK
4	John	Reinhardt	25	UK
5	Betty	Doe	28	UAE

Orders

order_id	item	amount	customer_id
1	Keyboard	400	4
2	Mouse	300	4
3	Monitor	12000	3
4	Keyboard	400	1
5	Mousepad	250	2

Shippings

shipping_id	status	customer
1	Pending	2

Top 10% query

```
select*
from customers
limit (select ceil (count(*)*0.1) from customers)
```

last 10%

```
select*
from customers
order by customer_id desc
limit (select ceil (count(*)*0.1) from customers)
```

Window Function

A window function is an analytic function in SQL that performs calculations across a set of rows related to the current row but does not reduce the number of rows.

2. ROW_NUMBER()

ROW_NUMBER() assigns a unique sequential number to each row within a partition, starting from

1.

 Each row gets a different number even if values are the same.

3. RANK()

RANK() assigns a rank to each row within a partition based on a column.

 Rows with the same value get the same rank, and the next rank skips numbers (gaps appear).

4. DENSE_RANK()

DENSE_RANK() is like RANK(), but the ranks are consecutive without gaps.

 Rows with the same value get the same rank.

5. MAX() as a Window Function

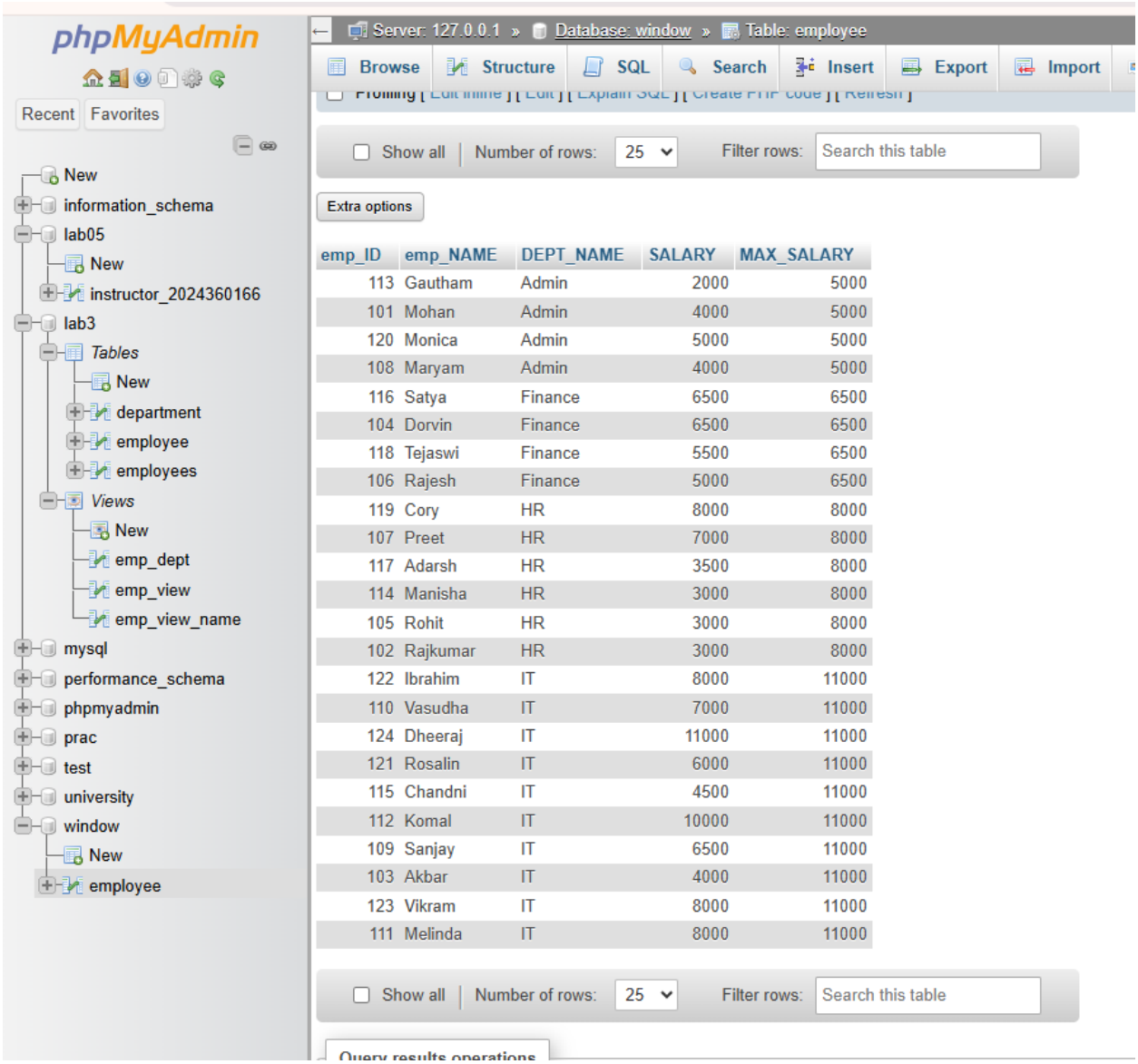
MAX() can be used as a window function to find the maximum value in a partition.

 Unlike aggregate MAX, it does not reduce the number of rows.

 Each row still appears and shows the maximum of its group.

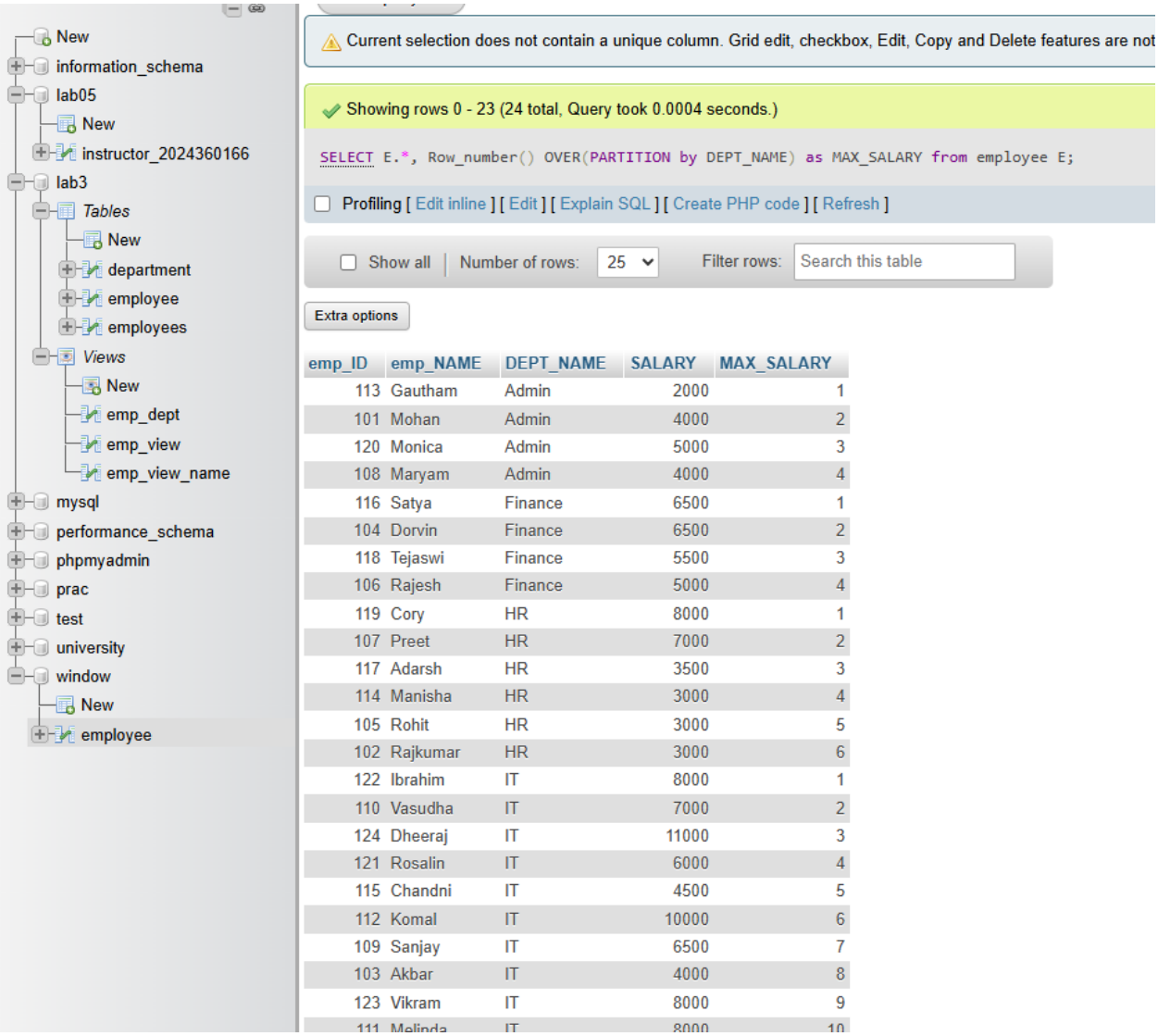
SELECT E.*, max(SALARY) OVER(PARTITION by DEPT_NAME) as MAX_SALARY

from employee E



SELECT E.*, Row_number() OVER(PARTITION by DEPT_NAME) as MAX_SALARY

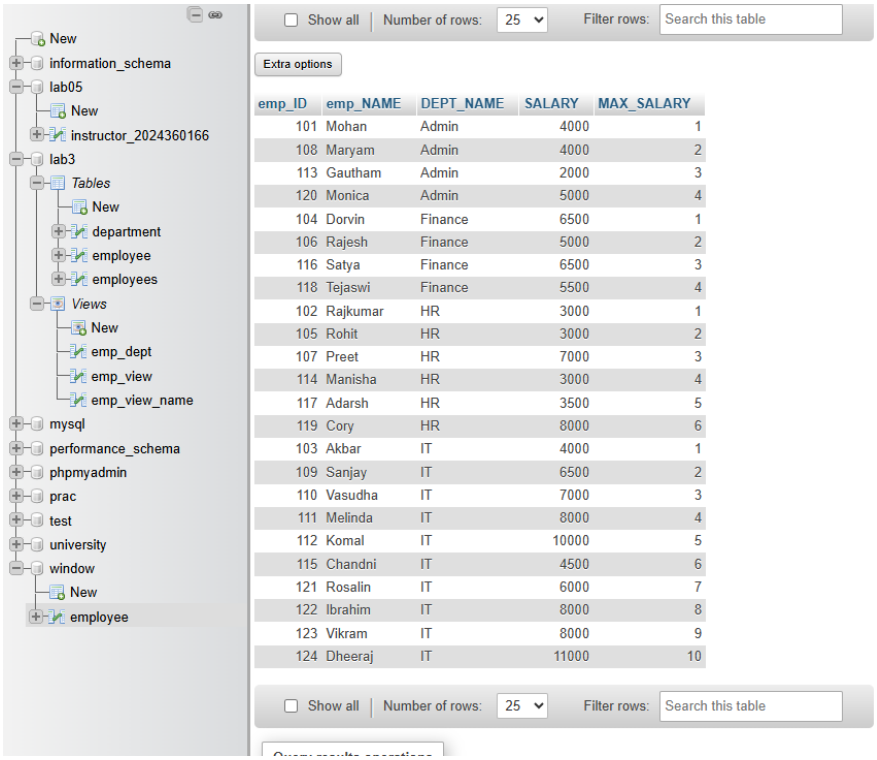
from employee E



emp_ID	emp_NAME	DEPT_NAME	SALARY	MAX_SALARY
113	Gautham	Admin	2000	1
101	Mohan	Admin	4000	2
120	Monica	Admin	5000	3
108	Maryam	Admin	4000	4
116	Satya	Finance	6500	1
104	Dorvin	Finance	6500	2
118	Tejaswi	Finance	5500	3
106	Rajesh	Finance	5000	4
119	Cory	HR	8000	1
107	Preet	HR	7000	2
117	Adarsh	HR	3500	3
114	Manisha	HR	3000	4
105	Rohit	HR	3000	5
102	Rajkumar	HR	3000	6
122	Ibrahim	IT	8000	1
110	Vasudha	IT	7000	2
124	Dheeraj	IT	11000	3
121	Rosalin	IT	6000	4
115	Chandni	IT	4500	5
112	Komal	IT	10000	6
109	Sanjay	IT	6500	7
103	Akbar	IT	4000	8
123	Vikram	IT	8000	9
111	Melinda	IT	8000	10

SELECT E.*, Row_number() OVER(PARTITION by DEPT_NAME ORDER BY emp_id ASC) as MAX_SALARY

from employee E



SELECT p.*

FROM (SELECT E.*, Row_number() OVER(PARTITION by DEPT_NAME ORDER BY emp_id ASC) as RN

from employee E) p

WHERE p.rn<=2

Extra options

emp_ID	emp_NAME	DEPT_NAME	SALARY	RN
101	Mohan	Admin	4000	1
108	Maryam	Admin	4000	2
104	Dorvin	Finance	6500	1
106	Rajesh	Finance	5000	2
102	Rajkumar	HR	3000	1
105	Rohit	HR	3000	2
103	Akbar	IT	4000	1
109	Sanjay	IT	6500	2

Show all | Number of rows: 25 Filter rows: Search this table

SELECT E.*, Rank() OVER(PARTITION by DEPT_NAME ORDER BY SALARY desc) as Rank

from employee E

Extra options

emp_ID	emp_NAME	DEPT_NAME	SALARY	Rank
120	Monica	Admin	5000	1
101	Mohan	Admin	4000	2
108	Maryam	Admin	4000	2
113	Gautham	Admin	2000	4
104	Dorvin	Finance	6500	1
116	Satya	Finance	6500	1
118	Tejaswi	Finance	5500	3
106	Rajesh	Finance	5000	4
119	Cory	HR	8000	1
107	Preet	HR	7000	2
117	Adarsh	HR	3500	3
114	Manisha	HR	3000	4
105	Rohit	HR	3000	4
102	Rajkumar	HR	3000	4
124	Dheeraj	IT	11000	1
112	Komal	IT	10000	2
123	Vikram	IT	8000	3
111	Melinda	IT	8000	3
122	Ibrahim	IT	8000	3
110	Vasudha	IT	7000	6
109	Sanjay	IT	6500	7
121	Rosalin	IT	6000	8
115	Chandni	IT	4500	9
103	Akbar	IT	4000	10

Show all | Number of rows: 25 Filter rows: Search this table

```
SELECT p.*

FROM  (SELECT E.*, Rank() OVER(PARTITION by DEPT_NAME ORDER BY SALARY desc) as Rank

from employee E)  p

WHERE p.rank<=3
```

emp_ID	emp_NAME	DEPT_NAME	SALARY	Rank
120	Monica	Admin	5000	1
108	Maryam	Admin	4000	2
101	Mohan	Admin	4000	2
116	Satya	Finance	6500	1
104	Dorvin	Finance	6500	1
118	Tejaswi	Finance	5500	3
119	Cory	HR	8000	1
107	Preet	HR	7000	2
117	Adarsh	HR	3500	3
124	Dheeraj	IT	11000	1
112	Komal	IT	10000	2
123	Vikram	IT	8000	3
111	Melinda	IT	8000	3
122	Ibrahim	IT	8000	3

```
SELECT E.*, dense_Rank() OVER(PARTITION by DEPT_NAME ORDER BY SALARY desc) as Rank

from employee E
```

Extra options

emp_ID	emp_NAME	DEPT_NAME	SALARY	Rank
120	Monica	Admin	5000	1
108	Maryam	Admin	4000	2
101	Mohan	Admin	4000	2
113	Gautham	Admin	2000	3
116	Satya	Finance	6500	1
104	Dorvin	Finance	6500	1
118	Tejaswi	Finance	5500	2
106	Rajesh	Finance	5000	3
119	Cory	HR	8000	1
107	Preet	HR	7000	2
117	Adarsh	HR	3500	3
114	Manisha	HR	3000	4
105	Rohit	HR	3000	4
102	Rajkumar	HR	3000	4
124	Dheeraj	IT	11000	1
112	Komal	IT	10000	2
123	Vikram	IT	8000	3
111	Melinda	IT	8000	3
122	Ibrahim	IT	8000	3
110	Vasudha	IT	7000	4
109	Sanjay	IT	6500	5
121	Rosalin	IT	6000	6
115	Chandni	IT	4500	7
103	Akbar	IT	4000	8